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Distant VCs and IPO performance

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ABSTRACT

This paper investigates the impact of institutional, cultural, and geographical distances between venture capital (VC) and entrepreneurial firms on the IPO performance of entrepreneurial firms. We propose that 'distant' VCs are more likely to suffer from information asymmetry, which may decrease their effectiveness in monitoring and adding value. This, in turn, could lead to higher uncertainty and more severe underpricing in entrepreneurial firms. Our analyses show greater institutional, cultural, and geographical distances are associated with higher IPO underpricing, while institutional and cultural distances have a greater impact than the more understood geographical distances. The adverse effects of distance are amplified for VCs with weaker investment networks and for IPO firms located in regions with a lower degree of marketization, where access to critical information and resources is constrained. Our main findings continue in a propensity score matching analysis, instrumental variables approach, and other robustness checks. Additionally, we provide evidence that these distances, particularly institutional distance, negatively impact post-IPO market performance, reinforcing the notion that various distances diminish VCs' effectiveness.

KEYWORDS

Institutional distance; cultural distance; geographical distance; venture capital; IPO performance

JEL CLASSIFICATION

G24; G30

I. Introduction

Venture capital (VC) plays a crucial role in nurturing entrepreneurial firms by providing advisory and monitoring services through industry experience and management expertise (e.g. Gompers and Lerner 2001; Hellmann and Puri 2002). As strategic investors, VCs help mitigate the risks faced by young firms, guiding them through key stages such as deal selection, business development, and ideally, eventual public offerings. However, the effectiveness of these services may be hindered when VCs are geographically, institutionally, and culturally distant from their investee firms. This conjecture aligns with the growing body of research suggesting that these distances between VCs and startups can constrain the VC's ability to effectively provide value-adding support (e.g. Li et al. 2014; Mingo, Morales, and Dau 2018).

While much of the existing literature has focused on the effects of these distances on VC market entry choice (e.g. Huang et al. 2023; Mingo, Morales, and Dau 2018; Nguyen et al. 2023), investment strategy (e.g. Chemmanur, Hull, and

Krishnan 2016; Dai and Nahata 2016; Tian 2011), and investment performance and exit outcomes (e.g. Li et al. 2014; Nahata, Hazarika, and Tandon 2014), little attention has been given to how these distances influence IPO performance. IPO success is largely contingent on investors' ability to mitigate uncertainty and provide value-adding oversight, understanding the impact of VC-investee distances on IPO underpricing – an outcome largely driven by information asymmetry – represents a critical gap in the literature that this paper seeks to address.

Drawing on information asymmetry theory, we posit institutional, cultural, and geographical distances elevate information costs and reduce the efficacy of deal selection, contracting, monitoring, and advisory functions of VCs, ultimately leading to greater IPO underpricing. Our analysis of 690 VC-backed IPOs listed in China provides compelling evidence that greater institutional, cultural, and geographical distances are associated with higher IPO underpricing. Interestingly, our findings reveal that institutional and cultural distances exert a more significant impact on IPO

underpricing than geographical distance, with the latter having traditionally been the focal point of IPO research. These results are robust to the instrumental variables approach and other checks including alternative measurements of IPO underpricing, and institutional and cultural distances.

To investigate the information asymmetry channel, we argue that the adverse effects of institutional, cultural, and geographical distances should be more pronounced when VCs or IPO firms face greater information limitations. Our findings support this hypothesis, demonstrating that the adverse effects of distances on IPO underpricing are more significant for VCs with weaker investment network centrality, as VCs with limited connections have difficulties reducing the uncertainty associated with distant investments. Similarly, the effect is stronger for IPO firms located in regions with a lower degree of marketization, where access to critical resources and information is more constrained.

Furthermore, using the propensity score matching method, we find that the influence of these distances is more likely attributable to the treatment effect – distant VCs exhibit comparatively weaker capacities for overseeing and fostering the growth of investee firms. Besides, we decompose the IPO underpricing into primary market underpricing and secondary market overvaluation. Our results suggest that institutional cultural, and geographical distances mainly intensify the IPO underpricing in the primary market, rather than the secondary market overvaluation.

Finally, the analyses of post-IPO market-adjusted BHARs reveal that these distances, particularly institutional distance, negatively affect post-IPO performance. This reinforces our explanations that these distances weaken VCs' supervisory and value-adding functions, contributing to IPO underpricing.

This paper makes several contributions. First, to the best of our knowledge, we are the first to examine the relationship between multi-dimensional VC-investee distances and IPO performance, complementing previous studies that have focused on the effects of various distances on VC investment decisions and exit outcomes (e.g. Chemmanur, Hull, and Krishnan 2016; Nahata, Hazarika, and Tandon 2014; Mingo,

Morales, and Dau 2018; Nguyen et al. 2023; Tian 2011). Second, our findings reveal that these three distance factors exacerbate IPO underpricing by aggravating information asymmetry and diminishing the effectiveness of VC value-adding activities. The results thus offer new insights into how spatial and contextual factors influence the IPO process. Our study thereby introduces a novel dimension to the traditional VC and IPO research, where the focus is on the resources and experience of VCs (e.g. Chahine et al. 2012; Shuwaikh, Dubocage, and Murer 2023; Tian 2012). Third, the significance of institutional and cultural distances over geographical distance complements the conventional focus on physical proximity in VC and IPO research (e.g. Cumming and Dai 2010; Tian 2011; F. Wang et al. 2022).

The rest of the paper is organized as follows. Section II presents the related literature and develops the main hypothesis. Section III describes our sample and defines all the key variables. In Section IV, we show our main results. In Section V, we conduct several robustness tests. Then, in Section VI we examine post-IPO market performance. Finally, Section VII concludes.

II. Related literature and hypothesis development

Previous research highlights VCs' crucial role in pinpointing promising ventures and providing strategic support (Hellmann and Puri 2002). VC-backed IPOs typically exhibit lower underpricing, attributed to VCs' quality screening, monitoring, and reputational certification. Barry et al. (1990) analyse U.S. IPO data from 1978 to 1987, linking reduced underpricing to venture capital's selective 'screening' of high-potential firms and the 'monitoring' that improves their management. Similarly, Megginson and Weiss (1991) also note a lower underpricing trend in VC-backed IPOs from 1983 to 1987, which they credit to the reputational certification effect of venture capital, where VCs' reputation is at stake, making their endorsement a mark of quality. Nevertheless, Gompers (1996) proposes the so-called *grandstanding hypothesis*, suggesting that young VCs might underprice IPOs to gain recognition and establish a favourable reputation. Chemmanur and

Loutskina (2006) propose the *market power hypothesis*, that VCs attract high-quality market participants, such as underwriters and institutional investors, to an IPO, achieving higher valuations for VC-backed firms.

Subsequent empirical studies offer mixed findings on the role of VCs in IPO underpricing. For instance, Tian (2012) and Chahine et al. (2012) emphasize the impact of VC syndicates with diverse skills, information, and networks in influencing IPO underpricing. While Tian (2012) finds that diversified VC syndicates reduce underpricing by providing valuable input to entrepreneurial firms, Chahine et al. (2012) suggest that principal-principal agency conflicts within VC syndicates may exacerbate underpricing. Additionally, Shuwaikh, Dubocage, and Murer (2023) report significantly lower underpricing in corporate VC-backed IPOs due to strategic support and long-term development promotion.

While much of this research is grounded in Western institutional contexts, an emerging body of literature explores IPO underpricing within China's distinctive regulatory and market environment. In China, the presence of politically connected VCs has been found to exacerbate underpricing by supporting less mature firms (R. Wang and Wu 2020). Hu et al. (2021) find that prestigious underwriters can mitigate underpricing, while Feng, Johansson, and Wei (2023) highlight the role of analyst attention in amplifying investor sentiment and increasing underpricing. S. Wang et al. (2023) demonstrate that boards with good reputations and extensive experience help reduce uncertainty and curb underpricing. Moreover, Zou et al. (2020), for instance, argue that underpricing may result more from investor irrationality rather than from deliberate issuer strategies.

Despite the insights offered by the existing literature, most studies have largely overlooked how spatial and relational frictions, such as institutional, cultural, and geographical distances, affect the effectiveness of VC support in IPO firms. Our study aims to fill this gap by investigating how these distances impact IPO underpricing in China's unique institutional context. We propose that these distances can compromise VCs' supervisory and managerial effectiveness, thus inversely affecting IPO performance.

Institutional distance involves variations in regulatory frameworks and business practices across countries or regions, requiring VCs to adjust their strategies to manage varying risks (Mingo, Morales, and Dau 2018; Peng 2003). In strong institutional settings, VCs can access reliable information to assess entrepreneurial potential and risks (Cumming, Schmidt, and Walz 2010). However, in weaker settings, information is often unreliable (Bruton and Ahlstrom 2003), and effective management practices and investor protection are also more challenging (Ahlstrom and Bruton 2006; Mingo, Morales, and Dau 2018).

Cultural distances in terms of values, business ethics, and behavioural norms among investment entities can hinder VC performance by disrupting effective communication and trust, both of which are essential for investment success (Dai, Jo, and Kasscieh 2012). Varied communication styles and patterns also complicate information transmission essential for VC operations (Mueller and Thomas 2001). Furthermore, cultural differences erode trust crucial for establishing cooperative investment ties, making it harder for VCs to effectively oversee and add value to investments (Bottazzi, Da Rin, and Hellmann 2016). Geographical distance, a more universally understood factor, is critical as well. Geographical proximity facilitates frequent interactions, enabling VCs to gather essential tacit knowledge and monitor operations effectively (Cumming and Dai 2010; Gupta and Sapienza 1992; Sorenson and Stuart 2001). Increased geographical distance hinders VCs' ability to provide timely advice and support. Therefore, our main hypothesis is as follows:

Hypothesis: The larger the institutional, cultural, and geographical distances between VCs and investee firms, the higher the level of IPO underpricing.

III. Data and variables

Data and sample

To investigate the impact of various distances between VCs and their investee companies on IPO outcomes, we assemble a dataset of companies

that have gone public with VC backing from 2007 to 2018 in China.¹ We sourced firm-specific IPO data from the China Stock Market & Accounting Research Database (CSMAR) database and venture capital-related data retrieved from the Zero2IPO Group, both are highly esteemed within the realm of research on Chinese corporate finance and the VC market.

Dependent variable

Following prior studies (e.g. Chen et al. 2020; Duong et al. 2022), we calculate IPO underpricing (*Underpricing*) as the first-day closing price of an IPO firm minus its offer price and divided by the offer price.

Independent variables

Institutional Distance

We measure institutional distance (*ID*) as the disparity in the institutional quality between the province where the VC firm's headquarter is located and the province where the investee firm is headquartered. To measure the quality of institutions in a province, we use a market development index, namely the Chinese Provincial Marketization Index (CPMI)² developed by the National Economic Research Institute (NERI) of China Reform Foundation (X. Wang, Fan, and Hu 2019). A higher CPMI score generally signifies a superior business and investment environment, marked by an effective government, minimal corruption, a mature capital market, and an equitable legal system. The CPMI is a well-established metric in scholarly work for gauging China's institutional landscape (e.g. R. Li, Ma, and Chen 2020; Yi, Wang, and Kafouros 2013).

Following the existing literature (e.g. Beugelsdijk, Ambos, and Nell 2018; Mingo, Morales, and Dau 2018), we compute the absolute difference of CPMI scores in the VC's and investee firm's provinces to measure the institutional distance (*ID*).³

Cultural Distance

We measure cultural distance (*CD*) using Zhao, Li, and Sun (2015) survey of Chinese provincial cultural values, which utilized GLOBE's Cultural Practices and Schwartz's Social Values questionnaires to assess regional cultural values based on first-year university students' hometowns.⁴ By aggregating the individual survey responses at the provincial level, they obtained cultural value data across 31 provinces. Their study is widely acknowledged in Chinese cultural research (e.g. S. Li, Gong, and Yang 2019; W. Liu and Huang 2019). We calculate the cultural distance (*CD*) utilizing the GLOBE cultural dimensions reported by Zhao, Li, and Sun (2015), applying the methodology established by Kogut and Singh (1988).

Geographical distance

To ascertain the geographical distance (*GD*) between the VC and the investee firm for each transaction, we first gather the latitude and longitude coordinates of their headquarter locations. Subsequently, employing spherical geometry, we apply the Haversine formula to compute the distances. It is delineated as follows:

$$\text{Geographical Distance} = R \times \arccos[\sin\beta_1\sin\beta_2 + \cos\beta_1\cos\beta_2\cos(\alpha_1 - \alpha_2)] \quad (1)$$

Here, R denotes the radius of the Earth, which is equal to 6,371 km. α_1 and β_1 denote the longitude and latitude of the headquarter city of a VC firm,

¹Due to data limitations on VC investments, our sample ends in 2018. To ensure consistency in the measurement of institutional and cultural distances, and to mitigate the confounding effects introduced by foreign VC, we exclude foreign VC involvement, both alone and in partnership with domestic VCs. Additionally, we omit IPO companies lacking essential data, resulting in a final sample of 690 VC-backed IPO firms for our study.

²The CPMI includes five components: (1) the relationship between government and market, mainly indicating government intervention and government efficiency; (2) the development of a non-SOE economy; (3) the development of product markets; (4) the development of factor markets; and (5) intermediary organization development and legal environment. These components measure different aspects of a province's marketization assessed by the NERI. The CPMI is the average of these five components. More details about the CPMI can be found in X. Wang, Fan, and Hu (2019).

³For multiple VC investors, we focus on the institutional distance of the lead VC to the investee firm, as lead VCs handle most due diligence and oversight activities. The lead VC is defined as the one with the largest cumulative investment or, in cases of a tie, the earliest investor (Nahata, Hazarika, and Tandon 2014). This approach also applies to measuring cultural and geographical distances. In an un-tabulated analysis, we define the lead VC investor as the one who participated in the most financing rounds and was also involved in the initial financing round of the IPO firm (Megginson and Weiss 1991; Sørensen 2007). If multiple lead VCs meet this criterion, we select the one with the highest total investment. The alternative definition yields similar results.

⁴The survey collected a total of 4,929 questionnaire responses, of which 3,690 were valid. The survey encompassed 56 universities across various provinces, municipalities, and autonomous regions (excluding Hong Kong, Macau, and Taiwan), including '985' and '211' universities, as well as ordinary universities and vocational colleges.

respectively; α_2 and β_2 denote the longitude and latitude of the investee headquarter city, respectively.

Control variables

Following prior research, particularly within the Chinese context (e.g. Chen et al. 2020; Çolak, Durnev, and Qian 2017; Ellul and Pagano 2006; Feng, Johansson, and Wei 2023; S. Wang et al. 2023; Zou et al. 2020), we control for variables to mitigate potential confounding factors influencing IPO underpricing. These include firm characteristics, corporate governance, issuance characteristics, and market conditions. (1) Firm characteristics: *Age*, the natural logarithm of the years between the company's inception and its IPO; *Size*, the natural logarithm of a company's total revenue in the year before the IPO; *Lev*, the ratio of total liabilities to total assets in the year before the IPO; *ROA*, the net profit divided by total assets in the year before the IPO; *B/M*, the ratio of book value of assets to market value at the listing time; *P/E*, the natural logarithm of the ratio of the IPO offer price to earnings per share in the year before the IPO, aiming to control for the impact of China's price-to-earnings ratio regulation policy on IPO pricing; *SOE*, an indicator variable that equals 1 if the IPO company is state-owned and 0 otherwise. (2) Corporate governance: *Board*, the number of directors at the time of IPO. *Duality*, an indicator variable that equals 1 if the CEO also serves as the Chairman and 0 otherwise. *Analyst*, the number of analyst reports issued during the period from the prospectus issuing date to the day prior to the first trading date of the IPO stock. (3) Issuance characteristics: *Offer Size*, the natural logarithm of the total funds raised by the IPO company, expressed in ten thousand yuan; *Underwriter*, a proxy for underwriter reputation, measured by the natural logarithm of the total underwriting fee of the main underwriter in that year, in thousand yuan. (4) Market conditions: *CSI 300 Return*, the monthly return of China Securities Index 300 in the month before the IPO. *Hot Market*, a binary variable that equals 1 if the number of IPOs in the month before the IPO falls into the top quartile over the sample period.

Syndication, an indicator variable that equals 1 when at least two VCs invested before the IPO. Additionally, we control for industry and year fixed effects and winsorize all continuous variables at the 1st and 99th percentiles to minimize the impact of outliers.

IV. Empirical results

Summary statistics

Table 1 Panel A reports the sample distribution in our study.⁵ We split the sample to check if the lead VC firm is located in the same province as the IPO firm. The results reveal that cross-provincial VC-backed IPOs (434) substantially outnumber local VC-backed IPOs (256), making up 62.9% of the total. This highlights the prevalence of cross-provincial VC investments in China, emphasizing the need to focus on their role. Furthermore, when analysing firms with only one VC investor, cross-provincial investments remain common. Among the 284 single-VC-backed IPO firms, 180 (63.4%) were backed by cross-provincial VCs, indicating that cross-provincial investment is prevalent even without multiple VC investors.

Panel B reports the descriptive statistics for our study's 690 observations of IPO firms from 2007 to 2018. Due to institutional distance data being available only between 2008 and 2016, the sample for this variable includes 675 observations. Descriptive statistics reveal that the average underpricing of VC-backed IPOs is 42.1%, ranging from -26.3% to 232.2%. The average institutional, cultural, and geographical distances between venture capital and firms are 0.927, 1.385, and 4.855, respectively, with significant variability in these distances across the sample, which facilitates the examination of their impact on IPO performance.

Since the correlation coefficients among the distance variables are relatively high in our sample (the correlation results are reported in Table A1), we introduce two additional distance transformations. Firstly, we compute the first principal component of all three distance variables (*DIST_PCA*), which aims to capture the primary influence

⁵Due to the suspension of IPOs throughout 2013 in China, combined with missing values for key variables, no observations are available for that year.

Table 1. Sample distribution and summary statistics.

Listing Year	Full Sample			Subsample (IPOs with only one VC investor)		
	Total	IPOs backed by cross-provincial VCs	IPOs backed by local VCs	Total	IPOs backed by cross-provincial VCs	IPOs backed by local VCs
Panel A: Sample distribution over time						
2007	2	2	0	2	2	0
2008	16	12	4	14	10	4
2009	22	8	14	12	3	9
2010	104	67	37	49	34	15
2011	87	56	31	37	27	10
2012	58	39	19	28	16	12
2014	57	33	24	28	18	10
2015	99	53	46	38	24	14
2016	81	49	32	28	15	13
2017	139	99	40	44	30	14
2018	25	16	9	4	1	3
Total	690	434	256	284	180	104
Panel B: Summary Statistics						
Variable	Obs.	Mean	Std.	Min.	P50	Max
Underpricing	690	0.421	0.288	-0.263	0.440	2.322
ID	675	0.927	1.308	0	0.45	8.91
CD	690	1.385	1.240	0	1.353	4.308
GD	690	4.855	2.933	0	6.361	8.148
Syndication	690	0.588	0.492	0	1	1
Age	690	2.491	0.483	0.693	2.565	3.584
Size	690	20.202	0.960	18.095	20.073	23.898
Lev	690	0.431	0.164	0.052	0.431	0.982
ROA	690	0.129	0.069	0.005	0.117	0.488
B/M	690	0.620	0.168	0.124	0.630	0.990
P/E	690	3.157	0.411	1.760	2.907	4.626
SOE	690	0.088	0.284	0	0	1
Board	690	8.587	1.376	5	9	15
Duality	690	0.178	0.383	0	0	1
Offer Size	690	10.821	0.685	8.966	10.753	13.123
Underwriter	690	12.613	1.125	9.105	12.825	14.927
Analyst	690	8.325	6.556	1	7	37
CSI 300 Return	690	0.012	0.068	-0.242	0.015	0.172
Hot Market	690	0.578	0.494	0	1	1

Panel A presents the distribution of VC-backed IPO firms listed during 2007–2018. The full sample consists of 690 IPO firms and the subsample (i.e. IPOs with only one VC investor) consists of 284 IPO firms. The main data sources are the CSMAR database and the Zero2IPO database. A VC is defined as a local VC if it is located in the same province as the invested company, otherwise a cross-provincial VC. In the full sample, if more than two VCs invest in a company, we only focus on the location of the lead VC. Panel B presents the descriptive statistics of the main variables for the 690 IPO firms in our sample. The number of observations is 675 for the institutional distance measure (ID) because of data availability.

exerted by all distance factors.⁶ Secondly, we derive the ‘orthogonal’ versions of the three distance variables, to evaluate the marginal impact of each specific distance dimension, decoupled from the effects of the other distance characteristics.⁷

Baseline regression analysis

We use the following baseline regression model to examine the impact of VC-invested firm distances on the IPO underpricing:

$$\text{Underpricing}_{i,t} = \beta_0 + \beta_1 \text{DIST}_{i,j,t-1} + Z_{i,t-1} + \mu_i + \gamma_t + \varepsilon_{i,t} \quad (2)$$

where i, j, t denote the IPO firm, lead VC, and IPO year, respectively. *Underpricing* is the dependent variable. *DIST* measures the distance variables. Z is the vector of control variables as previously defined. The model is estimated using ordinary least squares (OLS) with standard errors clustered by industry for heteroskedasticity.

⁶Principal component analysis (PCA), as introduced by Hotelling (1933), allows us to isolate the principal components of our three distance variables. PCA identifies the linear combination of the original distance variables that accounts for the largest possible variance, thus capturing as much of the data's variability as possible. This method enables us to discern the impact of the most significant ‘common element’ shared by all distance measures. The first principal component, a nearly equal-weighted linear combination of all three distance variables, accounts for 73% of the total variance.

⁷Orthogonalization is achieved using the modified Gram-Schmidt process (Golub and Van Loan 1996). This transforms our three independent distance variables into a mutually orthogonal set of transformed distance variables, preserving the original three-dimensional subspace. This allows us to capture the unique impact of each distance dimension, uncorrelated with the others. We use the order: institutional distance, cultural distance, geographical distance, but alternative orderings yield similar results.

Table 2. Distances and IPO underpricing: baseline regression results.

Dep. Var.: Underpricing	(1)	(2)	(3)	(4)	(5) Ortho
ID	0.018*** (0.006)				0.018** (0.007)
CD		0.020*** (0.006)			0.020** (0.008)
GD			0.004** (0.002)		0.010* (0.004)
DIST_PCA				0.014** (0.005)	
Syndication	-0.009 (0.007)	-0.000 (0.008)	-0.001 (0.009)	-0.009 (0.007)	-0.008 (0.007)
Age	0.008 (0.019)	-0.007 (0.020)	-0.008 (0.020)	0.005 (0.018)	0.011 (0.019)
Size	0.054** (0.019)	0.052** (0.018)	0.051** (0.018)	0.054** (0.019)	0.055** (0.019)
Lev	-0.193*** (0.044)	-0.157*** (0.050)	-0.167*** (0.051)	-0.193*** (0.047)	-0.188*** (0.048)
ROA	-0.566*** (0.116)	-0.476** (0.184)	-0.477** (0.182)	-0.571*** (0.128)	-0.583*** (0.123)
B/M	-0.473*** (0.107)	-0.474*** (0.103)	-0.467*** (0.106)	-0.477*** (0.108)	-0.478*** (0.104)
P/E	-0.094 (0.062)	-0.095 (0.058)	-0.098 (0.059)	-0.089 (0.061)	-0.095 (0.063)
SOE	0.085** (0.030)	0.085** (0.029)	0.085** (0.030)	0.089** (0.031)	0.084** (0.029)
Board	0.000 (0.004)	0.003 (0.004)	0.003 (0.004)	0.001 (0.004)	0.000 (0.004)
Duality	0.010 (0.010)	0.002 (0.012)	0.003 (0.012)	0.006 (0.011)	0.009 (0.011)
Offer Size	-0.082*** (0.018)	-0.085*** (0.016)	-0.085*** (0.017)	-0.081*** (0.017)	-0.081*** (0.017)
Underwriter	-0.012*** (0.004)	-0.009** (0.004)	-0.009* (0.004)	-0.012*** (0.003)	-0.012*** (0.004)
Analyst	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)
CSI 300 Return	0.942*** (0.082)	0.814*** (0.089)	0.812*** (0.093)	0.918*** (0.080)	0.937*** (0.080)
Hot Market	-0.059*** (0.016)	-0.054*** (0.016)	-0.053*** (0.017)	-0.057*** (0.016)	-0.059*** (0.016)
_cons	1.334*** (0.174)	2.504*** (0.434)	2.535*** (0.442)	1.354*** (0.177)	1.356*** (0.175)
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	675	690	690	675	675
Adjusted R-squared	0.263	0.305	0.300	0.262	0.265

This table presents the results from OLS regressions that investigate the relationship between institutional, cultural, and geographical distances and IPO underpricing, using a sample of VC-backed IPO firms in China from 2007 to 2018. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), and geographical distance (*GD*) in (3), the first principal component score in (4), and the orthogonal versions of the three distance measures in (5). The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table 2 presents the results of our analysis. Controlling for all influencing factors, *ID*, *CD*, and *GD* are statistically significant and positive at least at the 5% level in the first three columns, suggesting larger institutional, cultural, and geographical distances are related to more severe IPO underpricing, supporting our *Hypothesis*. The significantly positive coefficient for *DIST_PCA* in Column (4) implies that the three distance measures display a substantial joint common influence on the level of IPO underpricing. Meanwhile, Column (5) results

show a significant impact of orthogonally transformed institutional, cultural, and geographical distances on the IPO underpricing, suggesting these distances each retain a unique influence that is not captured by the other distance dimensions. Our findings also carry substantial economic significance. On average, a one-unit increase in institutional, cultural, and geographical distance is related to a 1.8%, 2%, and 0.4% increase in IPO underpricing, respectively.

Further, we standardize these variables to facilitate a comparative analysis of institutional,

Table 3. Standardized distances and IPO underpricing.

Dep. Var.: Underpricing	(1)	(2)	(3)
std_ID	0.024*** (0.008)		
std_CD		0.025*** (0.008)	
std_GD			0.011** (0.004)
Syndication	−0.009 (0.007)	−0.000 (0.008)	−0.001 (0.009)
Age	0.008 (0.019)	−0.007 (0.020)	−0.008 (0.020)
Size	0.054** (0.019)	0.052** (0.018)	0.051** (0.018)
Lev	−0.193*** (0.044)	−0.157*** (0.050)	−0.167*** (0.051)
ROA	−0.566*** (0.116)	−0.476** (0.184)	−0.477** (0.182)
B/M	−0.473*** (0.107)	−0.474*** (0.103)	−0.467*** (0.106)
P/E	−0.094 (0.062)	−0.095 (0.058)	−0.098 (0.059)
SOE	0.085** (0.030)	0.085** (0.029)	0.085** (0.030)
Board	0.000 (0.004)	0.003 (0.004)	0.003 (0.004)
Duality	0.010 (0.010)	0.002 (0.012)	0.003 (0.012)
Offer Size	−0.082*** (0.018)	−0.085*** (0.016)	−0.085*** (0.017)
Underwriter	−0.012*** (0.004)	−0.009** (0.004)	−0.009* (0.004)
Analyst	−0.002 (0.002)	−0.002 (0.002)	−0.002 (0.002)
CSI 300 Return	0.942*** (0.082)	0.814*** (0.089)	0.812*** (0.093)
Hot Market	−0.059*** (0.016)	−0.054*** (0.016)	−0.053*** (0.017)
_cons	1.351*** (0.177)	2.532*** (0.432)	2.554*** (0.441)
Industry	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	675	690	690
Adjusted R-squared	0.263	0.305	0.300

This table presents the results from OLS regressions that investigate the relationship between IPO underpricing and standardized distance factors, using a sample of VC-backed IPO firms in China from 2007 to 2018. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are standardized institutional distance (*std_ID*) in Column (1), standardized cultural distance (*std_CD*) in (2), and standardized geographical distance (*std_GD*) in (3). The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

cultural, and geographical distance impacts. The standardized distance variables, *std_ID*, *std_CD*, and *std_GD*, have zero mean and unit variance. We rerun the baseline regressions with the standardized distance variables and the results are reported in Table 3. As the table shows, IPO underpricing increases with distance factors. Cultural distance and institutional distance practically tie in the effect IPO underpricing, followed by institutional distance and geographical distance. This is in accord with relevant literature, suggesting that although geographical distance is commonly considered, institutional and cultural distances could

be more pivotal factors within the venture capital sphere (e.g. Colombo, d'Adda, and Quas 2019; Mingo, Morales, and Dau 2018; Tykvová and Schertler 2014).

Two-stage least squares regressions

Our analysis suggests that institutional, cultural, and geographical distances between VCs and IPO firms are positively associated with IPO underpricing. To address potential endogeneity concerns, we employ a two-stage least squares (2SLS) regression with instrumental variables. Specifically, we

use two instruments: *Hub VC*, an indicator variable that equals 1 if the VC is located in an innovation hub (i.e. Beijing, Shanghai, and Shenzhen), and *Cross-prov Exp*, the natural logarithm of the number of cross-province investments made by the VC. These variables are valid for several reasons. First, VCs based in major innovation centers are more likely to invest in firms across the country rather than restricting their investments to local markets.⁸ Second, VCs with more cross-province investment experience are both more inclined and better equipped to diversify their portfolios geographically. These factors make *Hub VC* and *Cross-prov Exp* strongly correlated with institutional, cultural,

and geographical distances between VCs and IPO firms. Importantly, these instruments do not directly affect IPO underpricing; rather, they influence distance-related monitoring effectiveness and information flow between VCs and their investee firms, which in turn impacts IPO underpricing. Furthermore, these instruments are exogenous because they are driven by historical and structural factors (such as the development of innovation ecosystems or the VC's past investment strategies) unrelated to the IPO pricing decisions.

Table 4 Panel A reports the first-stage results, showing that both *Hub VC* and *Cross-prov Exp* are highly significant and positively correlated with *ID*,

Table 4. Two-stage least squares regressions.

Dep. Var.:	(1) <i>ID</i>	(2) <i>CD</i>	(3) <i>GD</i>	(4) <i>DIST_PCA</i>
Panel A: First stage				
Hub VC	0.351*** (0.086)	0.887*** (0.133)	1.129*** (0.199)	0.783*** (0.136)
Cross-prov Exp	0.063* (0.034)	0.071** (0.028)	0.293*** (0.037)	0.123*** (0.030)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	675	690	690	675
Adjusted R-squared	0.046	0.148	0.104	0.118
Robust F-statistic	7.77	114.27	6.88	25.26
F-statistic p-value	0.00	0.00	0.00	0.00
Dep. Var.: Underpricing	(1)	(2)	(3)	(4)
Panel B: Second stage				
Instrumented ID	0.048** (0.024)			
Instrumented CD		0.039*** (0.011)		
Instrumented GD			0.018** (0.007)	
Instrumented DIST_PCA				0.032*** (0.012)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	675	690	690	675
Adjusted R-squared	0.285	0.338	0.320	0.295
Hansen J-statistic	1.30	1.31	3.56	1.65
p-Value	0.25	0.25	0.16	0.20

This table presents the results of 2SLS regression that investigates the relationship between distance factors and IPO underpricing based on the instrumental variable approach using *Hub VC* and *Cross-prov Exp* as the instruments for *ID*, *CD*, *GD*, and *DIST_PCA*. Panel A and Panel B report the first-stage and second-stage regression results, respectively. In Panel A, the dependent variables in Columns (1) - (4) are *ID*, *CD*, *GD*, and *DIST_PCA*, respectively. The F-statistic tests the null hypothesis of weak instruments, assessing whether the instruments are sufficiently correlated with the endogenous regressor to provide valid identification. In Panel B, the dependent variable is *Underpricing*. The Hansen J-test for overidentifying restrictions evaluates the joint null hypothesis that the excluded instruments are uncorrelated with the error term and are correctly excluded from the second-stage equation. All controls are the same with the baseline regression model of Equation (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

⁸In our sample, 533 VC firms are located in Beijing, Shanghai, or Shenzhen, while 250 are located outside these hub cities. Among the VCs located in hub cities, 75% of their invested IPO firms are located outside their home province. In contrast, only 32% of the IPO firms backed by VCs outside of hub cities are located in different provinces. This suggests that VCs in hub cities tend to invest in distant locations compared to their counterparts in non-hub cities.

CD, *GD* and *DIST_PCA*, indicating that they meet the relevance condition for valid instrumental variables. The first-stage *robust F-statistics* are 7.77, 114.27, 6.88 and 25.26, respectively, all significant at the 1% level, confirming that *Hub VC* and *Cross-prov Exp* are strong instruments. Panel B presents the second-stage results, where the instrumented distances remain positively and significantly associated with IPO underpricing, consistent with our main findings. The Hansen overidentification tests (*J-statistics*: 1.30, 1.31, 3.56 and 1.65, *p-values* >0.1) validate the instruments' exogeneity, confirming the robustness of our results.

Tests on information asymmetry channel

So far, the results indicate that distant VCs are positively associated with IPO underpricing, which can largely be attributed to weak monitoring capabilities resulting from information asymmetry. To test this conjecture, we hypothesize that the adverse effects of distances should be more pronounced when the VCs or the IPO firms face higher information asymmetry, i.e. when they have limited access to information.

We use three variables to capture the extent of information asymmetry. First, *Low Degree*, an indicator variable that equals 1 if the degree centrality of a VC is below its sample mean. Specifically, following Hochberg, Ljungqvist, and Lu (2007), we count the number of co-investments a VC has with other VCs based on its investment history. A higher degree of centrality indicates more opportunities for information and resource sharing, and vice versa (e.g. Sorensen and Stuart 2001; Hochberg, Ljungqvist, and Lu 2007). Second, we use the marketization index as a proxy for regional information asymmetry. Information economists like Verrecchia (1982) and Stiglitz (1989) have shown that information spreads more efficiently in highly market-oriented regions compared to those with lower levels of marketization. Based on this index, we construct two alternative indicator variables: *Less Developed*, which equals 1 if the IPO firm is located outside the top five provinces ranked by their average marketization index from 2008 to 2016 (i.e. Jiangsu, Shanghai, Zhejiang, Guangdong, and Beijing), and 0 otherwise, and *Low Marketization*, which equals 1 if the IPO

firm's provincial marketization index falls below the annual sample mean, and 0 otherwise.

The regression results, presented in Panel A, Panel B, and Panel C of Table 5, align with our predictions. Specifically, the interaction terms between distances and information asymmetry measures all exhibit positive and statistically significant coefficients, and the coefficients for *ID*, *CD*, and *GD* turn insignificantly negative. These findings further support our hypothesis that the effects of distance on underpricing are largely amplified (mitigated) when information access is more constrained (enhanced), highlighting the critical role of information transparency in the IPO process.

Treatment or selection effect?

Whereas the documented difference in the IPO underpricing appears to be due to the value-adding activities of different-distance VCs, distant-VC-backed and non-distant-VC-backed firms may differ in their characteristics. In this section, we study the impact of distance variables on IPO underpricing among the propensity score-matched pairs of entrepreneurial firms.

Specifically, the propensity scores are estimated based on a logit regression at the IPO firm level with the dependent variable, *High-dist*, which equals 1 if *DIST_PCA* is above its mean value and 0 otherwise. Independent variables are *Age*, *Size*, *Lev*, *ROA*, *B/M*, *SOE*, *Offer Size*, *Underwriter*, and industry and year effects. We subsequently execute a 1:1 nearest-neighbour propensity score matching without replacement employing the propensity score from the logit regression. This process resulted in 302 matched observations. We conduct diagnostic tests to assess the accuracy of the matching procedure. The univariate comparison between *High-dist* VC-backed and *Low-dist* VC-backed firms for the matched pairs shows insignificant differences between the two groups across all characteristics. Thus, the matching process removes meaningful differences along observable dimensions between these two groups of firms. These statistics are not tabulated for brevity.

Table 6 reports the analysis using the propensity score matched pairs. We find that after controlling for heterogeneous firm characteristics,

Table 5. Tests on information asymmetry channel.

<i>Dep. Var.: Underpricing</i>	(1)	(2)	(3)	(4)
Panel A: Low Degree as a proxy for information asymmetry				
ID	0.003 (0.011)			
ID × Low Degree	0.027** (0.011)			
CD		−0.001 (0.007)		
CD × Low Degree		0.042** (0.018)		
GD			−0.002 (0.002)	
GD × Low Degree			0.012** (0.005)	
DIST_PCA				−0.001 (0.007)
DIST_PCA × Low Degree				0.031** (0.011)
Low Degree	−0.012 (0.015)	−0.038** (0.015)	−0.037** (0.015)	0.017 (0.016)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	675	690	690	675
Adjusted R-squared	0.266	0.313	0.302	0.268
Panel B: Less Developed as a proxy for information asymmetry				
ID	0.010** (0.004)			
ID × Less Developed	0.026* (0.013)			
CD		0.003 (0.003)		
CD × Less Developed		0.061** (0.023)		
GD			0.002 (0.001)	
GD × Less Developed			0.013** (0.005)	
DIST_PCA				0.001 (0.002)
DIST_PCA × Less Developed				0.048*** (0.013)
Less Developed	−0.066** (0.025)	−0.108*** (0.034)	−0.087** (0.033)	−0.039** (0.015)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	675	690	690	675
Adjusted R-squared	0.267	0.317	0.300	0.273
Panel C: Low Marketization as a proxy for information asymmetry				
ID	0.010** (0.004)			
ID × Low Marketization	0.028* (0.015)			
CD		−0.001 (0.003)		
CD × Low Marketization		0.071*** (0.021)		
GD			0.001 (0.002)	
GD × Low Marketization			0.014** (0.005)	
DIST_PCA				0.002 (0.003)
DIST_PCA × Low Marketization				0.048*** (0.015)
Low Marketization	−0.069* (0.033)	−0.117*** (0.035)	−0.080* (0.038)	−0.040* (0.019)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	647	647	647	647
Adjusted R-squared	0.270	0.284	0.262	0.276

institutional, cultural, and geographical distances still have a significantly positive impact on IPO underpricing. Although we cannot completely rule out the selection effect, the propensity score matching suggests that this relationship is more likely to be attributed to the treatment effect. That is, distant VCs (i.e. with larger institutional, cultural, and geographical distances) are less able to monitor and nurture the investee firms.

Primary market underpricing or secondary market overvaluation?

The baseline analysis reveals that IPO underpricing depends on the IPO offer price in the primary market and the closing price in the secondary market. Since the pricing efficiency of the secondary market in China may be questionable, existing studies have found that Chinese A-share IPO underpricing includes both primary market underpricing and secondary market overvaluation (e.g. Gao 2010; Song, Tan, and Yi 2014).

To investigate whether the impact of distance factors on IPO underpricing manifests in the primary market or the secondary market, we adopt the methodology of Purnanandam and Swaminathan (2004) to decompose the IPO underpricing into primary market underpricing (*Underpricing_primary*) and secondary market overvaluation (*Underpricing_secondary*). The procedure is briefly stated as follows:

From the CSMAR database, we select groups of companies in the ‘same industry – similar sales – similar EBITDA’ category as the IPO firm, we then choose the company with sales closest to that of the IPO firm as the benchmark company. Following automated matching and subsequent manual checks, we secure data for 448 companies comparable to the IPO firms.

Based on this, we estimate IPO firms’ intrinsic value (*IV*) as:

$$IV = \left(\frac{P}{\text{Earnings/Shares}} \right)_{\text{Match}} \times (\text{Earnings/Shares})_{\text{IPO}} \quad (3)$$

where

$$\begin{aligned} & \left(\frac{P}{\text{Earnings/Shares}} \right)_{\text{Match}} \\ &= \frac{\text{Closing price on the listing day of the comparable company}}{(\text{Net profit/Total shares}) \text{ at the year before IPO of the comparable company}} \end{aligned} \quad (4)$$

$$(\text{Earnings/Shares})_{\text{IPO}} = \frac{\text{Net profit at the year before IPO of the IPO firm}}{\text{Total Shares of the IPO firm}} \quad (5)$$

Based on the *IV* of the IPO firm, the primary market underpricing and secondary market overvaluation are calculated as follows:

$$\text{Underpricing_primary} = \frac{IV}{\text{OfferPrice}} \quad (6)$$

$$\text{Underpricing_secondary} = \frac{\text{ClosingPrice}}{IV} \quad (7)$$

where, *Offer Price* is the offer price of the IPO firm, and *Closing Price* is the closing price of the IPO firm on the first day of listing.

We then rerun the baseline regression model with *Underpricing_primary* and *Underpricing_secondary* as the dependent variables and the results are reported in Panel A and Panel B of Table 7, respectively. Panel A shows that, institutional, cultural and geographical distances are positive and statistically significant at least at the 5% level. Panel B shows that the relation between secondary market overvaluation (*Underpricing_secondary*) and institutional, cultural, and geographical distances are not significant.

Through the breakdown of IPO underpricing into two components, our research reveals that in

This table presents the results from OLS regressions that investigate the information asymmetry channel through which institutional, cultural, and geographical distances affect IPO underpricing. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The variables of interest are the interaction terms between distances with information asymmetry variables. In Panel A, Panel B, and Panel C, we use *Low Degree*, *Less Developed*, and *Low Marketization* as proxies for information asymmetry, respectively. *Low Degree* is an indicator variable that equals 1 if the degree centrality of a VC is below its sample mean. *Less Developed* is an indicator variable that equals 1 if the IPO firm is located outside the top five provinces ranked by their average marketization index from 2008 to 2016 (i.e. Jiangsu, Shanghai, Zhejiang, Guangdong, and Beijing); *Low Marketization*, which equals 1 if the IPO firm’s provincial marketization index falls below the annual sample mean. All controls are the same with the baseline regression model of Equation (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table 6. Propensity score matching results.

	(1)	(2)	(3)	(4)	(5)
<i>Dep. Var.: Underpricing</i>					Ortho
ID	0.029*** (0.009)				0.031*** (0.007)
CD		0.025** (0.009)			0.026** (0.010)
GD			0.004** (0.002)		0.012* (0.006)
DIST_PCA				0.019** (0.006)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	302	302	302	302	302
Adjusted R-squared	0.235	0.235	0.224	0.232	0.239

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing, using a propensity score matched sample of VC-backed IPO firms in China from 2007 to 2018. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). All controls are the same with the baseline regression model of Equation (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table 7. IPO primary and secondary underpricing.

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Dep. Var.: Underpricing_primary</i>					Ortho
ID	0.181*** (0.049)				0.136*** (0.039)
CD		0.200*** (0.041)			0.159** (0.058)
GD			0.063** (0.025)		0.203** (0.072)
DIST_PCA				0.184*** (0.045)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	438	448	448	438	438
Adjusted R-squared	0.031	0.034	0.028	0.034	0.031
<i>Panel B: Dep. Var.: Underpricing_secondary</i>					
ID	-0.082 (0.050)				-0.132 (0.086)
CD		0.015 (0.028)			0.014 (0.036)
GD			-0.001 (0.019)		0.004 (0.064)
DIST_PCA				-0.020 (0.029)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	438	448	448	438	438
Adjusted R-squared	0.119	0.110	0.110	0.114	0.117

This table presents the results from OLS regressions that investigate the relationship between distance factors and decomposed IPO underpricing measures. In Panel A, the dependent variable is the primary market underpricing *Underpricing_primary*, intrinsic value divided by the offer price of IPO firms. In Panel B, the dependent variable is the secondary market overvaluation *Underpricing_secondary*, closing price on the first trading day divided by the intrinsic value of IPO firms. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). All controls are the same with the baseline regression model of Equation (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

China, the institutional, cultural, and geographical distances between venture capitalists and their investee firms predominantly exacerbate underpricing in the primary market.

V. Robustness checks

Controlling for government regulation

To curb speculative activities in the IPO market and improve stock market quality, the Shanghai and Shenzhen Stock Exchanges implemented a first-trading-day price limit policy on 13 December 2013, capping the declared price at no more than 144% of the offer price. To account for this regulatory alteration, we proceed to exclude companies that went public after this date. The results from re-estimating the baseline model with the subsample are presented in Appendix Table A2. The results indicate that the first-day price limit policy in 2013 did not materially alter our main conclusions.

Analysis of cross-provincial VC-backed IPOs only

In our sample, approximately 37% of IPO firms are backed by local VC investors. Although we consider zero values for institutional, cultural, or geographical distances essential for explaining the distance effect on IPO underpricing, as they capture the full range of these distances, we also conduct a robustness check by focusing on cross-provincial VC-backed IPOs to ensure that our results are not driven by the high proportion of local VC-backed firms. The regression results based on this subsample, reported in Appendix Table A3, remain qualitatively consistent with our main findings.

Controlling for underwriter proximity

Underwriters play a critical role in the IPO process (Carter, Dark, and Singh 1998). While we control for underwriter reputation in the baseline regression, there may be concerns that the location of the underwriter could confound our results. To address this

concern, we introduce underwriter proximity as an additional control. We use an indicator variable, *Local Underwriter*, to indicate whether the underwriter is based in the same city as the IPO firm. The regression results, with *Local Underwriter* included as an additional control, are reported in Appendix Table A4. The results reinforce our main finding that, even after accounting for underwriter proximity, distant VCs are associated with greater IPO underpricing.

Alternative measures of IPO underpricing

We construct an adjusted IPO underpricing variable, *Underpricing_Adj*, which is the first-day IPO return minus the market return on the listing day, to eliminate the potential interference of market conditions. Following Ellul and Pagano (2006), we also use $\ln(\text{Closing/Offer})$, the natural logarithm of the ratio of the closing price on the first trading day to the offer price, as an alternative measure. The regression results for these two dependent variables in Appendix Table A5 correspond with our main findings.

Alternative measures of institutional and cultural distances

In this section, we consider alternative measures of institutional and cultural distances. First, we use the 'intermediary organization development and legal environment' component of the Chinese Provincial Marketization Index (CPMI) to calculate institutional distance (*LegalDist*). Second, while the baseline regression treats institutional distance as non-directional, we also consider the directionality of institutional distance by calculating it as the difference between the CPMI scores of the lead VC's province and the IPO firm's province (*ID_D*). Third, we alternatively use Schwartz's cultural values scores in Zhao, Li, and Sun (2015) to calculate cultural distance between provinces (*CD_S*). Fourth, we follow Y. Liu, Jiao, and Xu (2020) to calculate the dialect distance between cities of VCs and IPO firms (*DialectDist*) as a proxy for cultural distance.⁹ The

⁹Specifically, this method utilizes the Chinese Dialect Family Tree from the *Language Atlas of China* (Chinese Academy of Social Sciences, and Australian Academy of the Humanities 1987) and the *Dictionary of Chinese Dialect* (Xu and Ichiro 1999). Chinese dialects are classified into ten dialectal super groups, namely, Mandarin, Wu, Gan, Xiang, Min, Hakka, Yue, Jin, Ping, and Hui. Each super group can be further divided into several smaller, i.e. second-tier dialectal groups, based on tonal categories and pitch contour. The smallest dialectal unit is the dialect sub-group. Each county (the smallest administrative division) typically has a unique dialect. Following the methodology of Y. Liu, Jiao, and Xu (2020), we calculate dialect distance between cities, which ranges from 0 to 3, with higher values indicating greater dialect distance between two cities.

results reported in Appendix Table A6 align with our baseline findings.

VI. Distances and post-IPO market performance

Previous research has consistently found that IPO firms typically exhibit poor long-run performance (Carter, Dark, and Singh 1998; Loughran, Ritter, and Rydqvist 1994; Ritter 1991). Information asymmetry and agency problems are critical factors contributing to this underperformance. In this section, we further analyse whether distant VCs weaken the

supervisory and value-adding functions, which result in poorer post-IPO performance.

Following Çolak, Durnev, and Qian (2017), we calculate the market-adjusted buy-and-hold abnormal return (BHAR) using Eq. (8) to measure the market performance of IPO firms.

$$BHAR_{i,t} = \prod_{t=1}^T (1 + r_{i,t}) - \prod_{t=1}^T (1 + r_{m,t}) \quad (8)$$

here $r_{i,t}$ represents the stock return in month t after the IPO, and $r_{m,t}$ represents the value-

Table 8. Distances and post-IPO performance.

	(1)	(2)	(3)	(4)
Panel A: Dep. Var.: BHAR1				
ID	−0.029 (0.020)			
CD		−0.002 (0.017)		
GD			−0.012 (0.010)	
DIST_PCA				−0.022 (0.018)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	661	676	676	661
Adjusted R-squared	0.352	0.341	0.343	0.352
Panel B: Dep. Var.: BHAR2				
ID	−0.042*** (0.013)			
CD		−0.002 (0.024)		
GD			−0.016 (0.012)	
DIST_PCA				−0.029 (0.020)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	661	676	676	661
Adjusted R-squared	0.251	0.256	0.259	0.250
Panel C: Dep. Var.: BHAR3				
ID	−0.075*** (0.010)			
CD		−0.012 (0.020)		
GD			−0.017 (0.010)	
DIST_PCA				−0.042** (0.015)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	661	676	676	661
Adjusted R-squared	0.158	0.163	0.164	0.155

This table presents the results from OLS regressions that investigate the relationship between distance factors and post-IPO performance. The dependent variables in Panels A, B, and C are the 12-month, 24-month, and 36-month market-adjusted buy-and-hold abnormal returns (BHAR), respectively. The distances between VCs and IPO firms are the institutional distance (ID) in Column (1), cultural distance (CD) in (2), geographical distance (GD) in (3), and the first principal component score of the three distance variables in (4). All controls are the same with the baseline regression model of Equation (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

weighted return of the total market capitalization of the Shanghai and Shenzhen A-shares for the corresponding period. We calculate the BHAR for 12 months, 24 months, and 36 months post-IPO, denoted as *BHAR1*, *BHAR2*, and *BHAR3*, respectively.

Panels A, B, and C of [Table 8](#) present the regression results with *BHAR1*, *BHAR2*, and *BHAR3* as the dependent variables, respectively. It can be observed that all three distance variables reduce the buy-and-hold abnormal returns for one-year, two-year, and three-year periods post-IPO, although only the effect of institutional distance is statistically significant. This finding reflects the persistent nature of institutional distance, suggesting that VCs seem to have larger difficulties to overcome institutional distance compared with cultural distance or geographical distance. This aligns with existing literature (e.g. Tykvová and Schertler 2014, Nahata, Hazarika, and Tandon 2014). Differences in regulatory quality, investor protection, contract enforcement, and market practices hinder the transferability of VC expertise and impede the effectiveness of VC oversight, even after the firm goes public. Unlike cultural or geographical distances, these institutional frictions are not easily neutralized by increased disclosure or market learning.

VII. Conclusions

Leveraging a dataset of 690 Chinese VC-backed IPOs from 2007 to 2018, we find a positive relation between institutional, cultural, and geographical distances and IPO underpricing. These results are consistently robust across various tests, including instrumental variables approach and alternative measures. Notably, the adverse impact of distances is more pronounced for VCs with weaker network ties and for IPO firms located outside major financial centres, highlighting how these distances exacerbate information asymmetry and uncertainty, which in turn influence IPO underpricing.

Propensity score matching confirm our results after accounting for selection bias. Moreover, our research reveals that the distances between venture capitalists and their investee firms predominantly exacerbate underpricing in the primary market, not the

secondary market overvaluation. Finally, our examination of post-IPO performance suggests that distances, especially institutional distance, negatively influence post-IPO market performance, reinforcing the notion that these distances can obstruct the value-adding contributions of venture capital.

This paper contributes to the literature by complementing previous studies that have focused on the effects of various distances on VC investment decisions and exit outcomes (e.g. Mingo, Morales, and Dau 2018; Nahata, Hazarika, and Tandon 2014; Tian 2011). Additionally, our findings offer new insights into how spatial and contextual factors influence the IPO process, thereby introducing a novel dimension to the traditional VC and IPO research, where the focus is on the resources and experience of VCs (e.g. Chahine et al. 2012; Shuwaikh, Dubocage, and Murer 2023). Practically, this study highlights the importance for both VCs and entrepreneurial firms to consider not only geographical proximity but also institutional and cultural contexts when assessing potential investments or selecting partners, as the value-adding role of VCs may be diminished in distant relationships, affecting IPO and post-IPO performance.

Our analysis raises several avenues for further research. One area worth investigating further is to explore the effect of these distances on IPO underpricing in other countries as well as in the context of cross-border VC investment. While our study draws on the regional disparities within China as an analogy for cross-border contexts, future research could investigate whether our findings hold true in other markets and further explore the effect of the cross-border VC involvement. Another important area worth investigating is to explore mechanisms that may help distant VCs overcoming various forms of distances. Different types of distances may require distinct mechanisms for VCs to effectively manage these challenges.

Disclosure statement

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Appendix

Table A1. Variable correlations.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
(1)	Underpricing	1									
(2)	ID	0.06	1								
(3)	CD	0.09	0.49	1							
(4)	GD	0.03	0.52	0.73	1						
(5)	Syndication	−0.03	0.00	−0.02	−0.01	1					
(6)	Age	−0.08	0.03	0.01	0.05	0.07	1				
(7)	Size	−0.09	0.06	−0.01	0.06	0.03	0.07	1			
(8)	Lev	0.00	−0.01	−0.06	0.00	0.02	−0.08	0.48	1		
(9)	ROA	−0.13	−0.07	0.04	−0.03	−0.15	−0.16	−0.36	−0.57	1	
(10)	B/M	−0.22	0.04	0.03	0.09	−0.04	−0.14	0.43	0.35	−0.13	
(11)	P/E	−0.07	−0.05	−0.05	−0.05	−0.03	−0.41	−0.21	0.07	0.13	
(12)	SOE	0.14	0.04	−0.03	−0.04	−0.09	−0.04	0.19	0.12	−0.16	
(13)	Board	0.06	0.07	−0.02	0.00	0.00	−0.10	0.18	0.14	−0.16	
(14)	Duality	−0.03	−0.05	0.05	0.04	0.07	0.09	0.03	−0.08	0.03	
(15)	Offer Size	−0.22	0.05	0.01	0.06	−0.03	−0.21	0.53	0.18	0.20	
(16)	Underwriter	−0.12	−0.01	0.05	0.06	0.04	−0.06	0.12	0.02	0.03	
(17)	Analyst	−0.08	−0.04	0.00	0.02	−0.08	−0.29	0.02	0.10	0.23	
(18)	CSI 300 Return	0.10	−0.05	−0.05	−0.05	0.02	0.12	0.02	−0.03	−0.04	
(19)	Hot Market	−0.08	0.05	0.05	0.08	0.10	0.02	0.04	0.06	−0.08	
	VIF	−	1.55	2.37	2.52	1.14	1.57	4.26	2.2	2.79	
(Continued)											
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)
(10)	B/M	1									
(11)	P/E	0.22	1								
(12)	SOE	0.09	0.02	1							
(13)	Board	0.17	0.08	0.24	1						
(14)	Duality	−0.06	−0.16	−0.15	−0.12	1					
(15)	Offer Size	0.44	0.39	0.13	0.14	−0.02	1				
(16)	Underwriter	0.10	0.13	−0.02	−0.01	0.08	0.22	1			
(17)	Analyst	0.14	0.42	0.02	0.04	−0.18	0.34	0.06	1		
(18)	CSI 300 Return	−0.22	−0.08	0.06	0.03	0.07	−0.04	0.004	−0.16	1	
(19)	Hot Market	−0.05	0.14	−0.05	0.02	0.13	0.14	0.26	−0.03	0.111	1
	VIF	3.65	4.55	1.25	1.21	1.21	3.85	1.25	2.14	1.28	2.29

This table presents the correlation coefficients among the main variables in our sample. Besides, the VIF values are reported at the bottom of this table.

Table A2. Robustness checks: controlling for the IPO regulation policy in China.

Dep. Var.: Underpricing	(1)	(2)	(3)	(4)	(5)
ID	0.039** (0.014)				Ortho 0.051** (0.017)
CD		0.024* (0.011)			0.016** (0.006)
GD			0.004 (0.004)		0.010 (0.010)
DIST_PCA				0.018** (0.008)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	274	288	288	274	274
Adjusted R-squared	0.445	0.450	0.447	0.437	0.443

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing using a subsample of VC-backed IPO firms in China from 2007 to 2013, considering the first trading-day price limit policies in China on December 13, 2013. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). All controls are the same with the baseline regression model of Eq (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table A3. Robustness checks: analysis of cross-provincial VC-backed IPOs only.

	(1)	(2)	(3)	(4)	(5)
<i>Dep. Var.: Underpricing</i>					Ortho
ID	0.019** (0.007)				0.020** (0.007)
CD		0.042** (0.016)			0.036*** (0.012)
GD			0.010** (0.004)		0.042 (0.035)
DIST_PCA				0.051*** (0.015)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	419	434	434	419	419
Adjusted R-squared	0.267	0.323	0.328	0.276	0.275

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing using a subsample of VC-backed IPO firms backed by cross-provincial VCs. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). All controls are the same with the baseline regression model of Eq (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table A4. Robustness checks: controlling for underwriter proximity.

	(1)	(2)	(3)	(4)	(5)
<i>Dep. Var.: Underpricing</i>					Ortho
ID	0.018*** (0.006)				0.018** (0.007)
CD		0.020*** (0.006)			0.020** (0.008)
GD			0.004** (0.001)		0.009* (0.004)
DIST_PCA				0.015** (0.005)	
Local Underwriter	−0.001 (0.015)	−0.008 (0.016)	−0.008 (0.017)	0.002 (0.015)	−0.008 (0.015)
Other Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	675	690	690	675	675
Adjusted R-squared	0.262	0.304	0.299	0.261	0.264

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing with underwriter proximity as an additional control. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). *Local Underwriter* is an indicator variable that equals 1 if the underwriter is based in the same city as the IPO firm. All other controls are the same as the baseline regression model of Eq (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table A5. Robustness checks: alternative IPO underpricing measures.

	(1)	(2)	(3)	(4)	(5)
					Ortho
Panel A: Dep. Var.: Underpricing_Adj					
ID	0.017** (0.007)				0.019** (0.008)
CD		0.017*** (0.005)			0.018** (0.008)
GD			0.003** (0.001)		0.008* (0.004)
DIST_PCA				0.013** (0.005)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	675	690	690	675	675
Adjusted R-squared	0.229	0.293	0.288	0.226	0.230
Panel B: Dep. Var.: Ln(Closing/Offer)					
ID	0.011** (0.004)				0.012** (0.005)
CD		0.011** (0.004)			0.012*** (0.004)
GD			0.002* (0.001)		0.004 (0.003)
DIST_PCA				0.008** (0.003)	
Controls	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	675	690	690	675	675
Adjusted R-squared	0.321	0.344	0.340	0.319	0.322

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing, using alternative underpricing measures. In Panel A, the dependent variable is *Underpricing_Adj*, the first-day return of the IPO minus the market return on the listing day. In Panel B, the dependent variable is *Ln(Closing/Offer)*, the natural logarithm of the ratio of the closing price on the first trading day to the offer price. The distances between VCs and IPO firms are the institutional distance (*ID*) in Column (1), cultural distance (*CD*) in (2), geographical distance (*GD*) in (3), the first principal component score of the three distance variables in (4), and the orthogonal versions of the three distance measures in (5). All controls are the same with the baseline regression model of Eq (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table A6. Robustness checks: alternative distance measures.

Dep. Var.: Underpricing	(1)	(2)	(3)	(4)
LegalDist	0.008** (0.003)			
ID_D		0.014** (0.005)		
CD_S			0.011** (0.004)	
DialectDist				0.006** (0.002)
Controls	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	647	673	688	686
Adjusted R-squared	0.265	0.262	0.303	0.300

This table presents the results from OLS regressions that investigate the relationship between distance factors and IPO underpricing, using alternative distance measures. The dependent variable is *Underpricing*, the first-day closing price of an IPO firm minus its offer price and then divided by the offer price. The distances between VCs and IPO firms are institutional distance in terms of legal and regulations (*LegalDist*) in Column (1), institutional distance with direction (*ID_D*) in (2), cultural distance calculated with Schwartz's cultural values (*CD_S*) in (3), and cultural distance calculated according to dialect distances (*DialectDist*) in (4). All controls are the same with the baseline regression model of Eq (2) and their estimated coefficients are not reported for brevity. The standard errors clustered by industry for heteroskedasticity are reported in parentheses. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.